

WILP, BITS Pilani

Session 12: Multiple Regression & Intro to Time Series

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Multiple & Non-Linear Regression

Extend simple lines to multiple predictors and curved non-linear patterns.

Multiple linear regression models a continuous target variable Y using multiple explanatory variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$. Non-linear models apply transformations like logarithms or quadratic terms when data follows curves.

Everyday Picture & Intuition

Predicting house price is not just about square footage. You also need bedrooms, neighborhood age, and distance to public transport. Multiple regression assigns weights to all predictors simultaneously.

1.1 Non-Linear Transformations

For diminishing returns, logarithmic transformations $Y = \beta_0 + \beta_1 \ln(X)$ or quadratic forms $Y = \beta_0 + \beta_1 X + \beta_2 X^2$ fit non-linear curved trends while remaining linear in parameters.

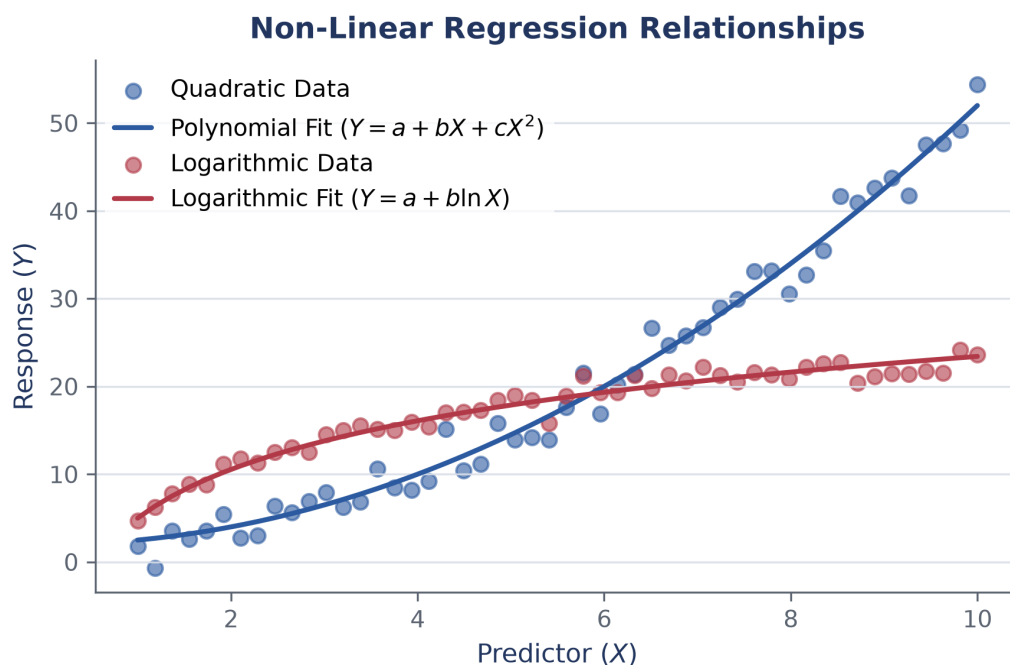


Figure 1: Polynomial and logarithmic regression curves fitting non-linear relationships.

Fully Worked Example

Consider fitting a quadratic regression $Y = 5.0 + 2.0X - 0.1X^2$ to estimate yield (Y) based on fertilizer dosage (X).

Step 1: Evaluate predicted yield at $X = 5$:

$$\hat{Y} = 5.0 + 2.0(5) - 0.1(5^2) = 5.0 + 10.0 - 2.5 = 12.5 \quad (1)$$

Step 2: Evaluate predicted yield at $X = 10$:

$$\hat{Y} = 5.0 + 2.0(10) - 0.1(10^2) = 5.0 + 20.0 - 10.0 = 15.0 \quad (2)$$

Notice how doubling X from 5 to 10 does not double the yield due to diminishing returns.

2 Time Series Components

Decompose historical sequence observations into structural components.

A time series is a set of quantitative observations ordered over successive periods. It breaks down into four primary components: Secular Trend (T_t), Seasonal variation (S_t), Cyclical variation (C_t), and Irregular noise (I_t).

Everyday Picture & Intuition

Think of sales data like ocean waves. The tide rising over years is the trend. Summer beach surges are seasonality. Multi-year economic booms are cycles. Random unexpected storm days are irregular noise.

2.1 Additive vs. Multiplicative Models

- **Additive Model:** $y_t = T_t + S_t + C_t + I_t$ (used when seasonal swings remain constant in dollar magnitude).
- **Multiplicative Model:** $y_t = T_t \times S_t \times C_t \times I_t$ (used when seasonal swings grow proportionally with overall trend level).

3 Stationarity and the Split-Half Test

Check whether time series statistical properties remain constant over time.

A time series is stationary if its mean, variance, and autocorrelation structure do not depend on time. Non-stationary series must be transformed (e.g., via first differencing $w_t = y_t - y_{t-1}$) prior to modeling.

Everyday Picture & Intuition

A stationary process is like a steady thermostat reading around 72 degrees. A non-stationary process is like a room heating up over time where the average temperature keeps rising.

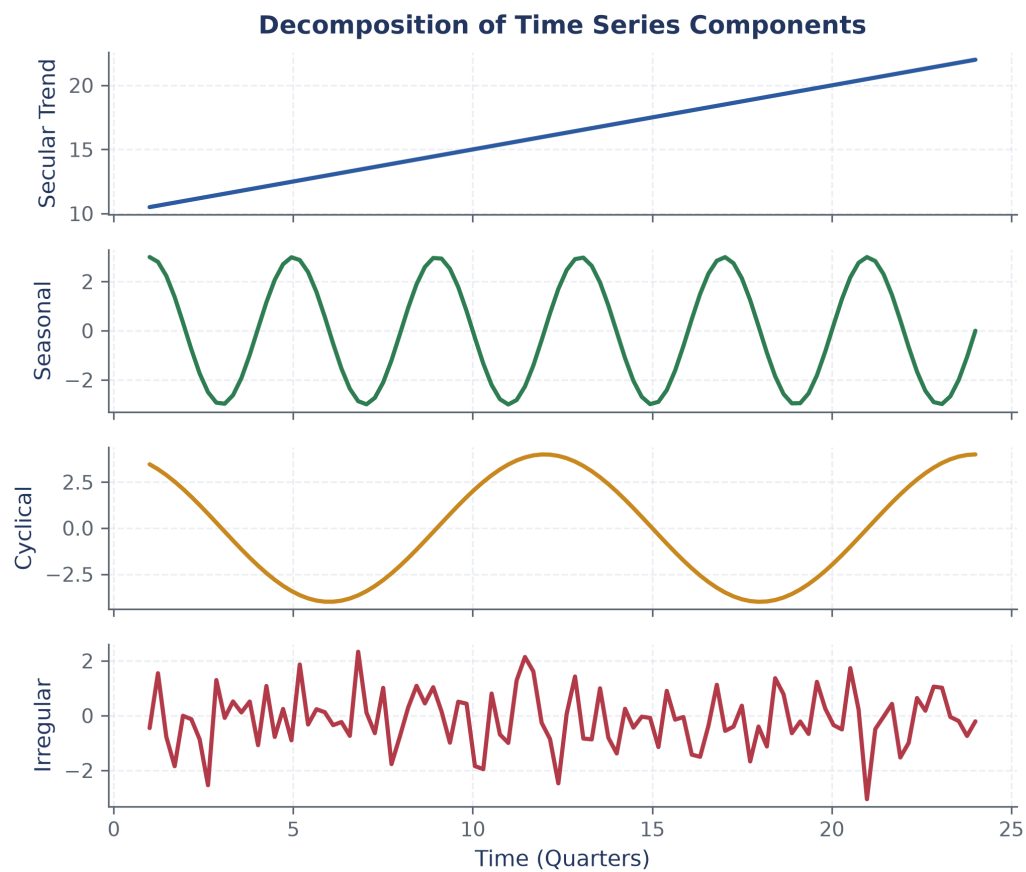


Figure 2: Decomposition of time series into trend, seasonal, cyclical, and irregular noise components.

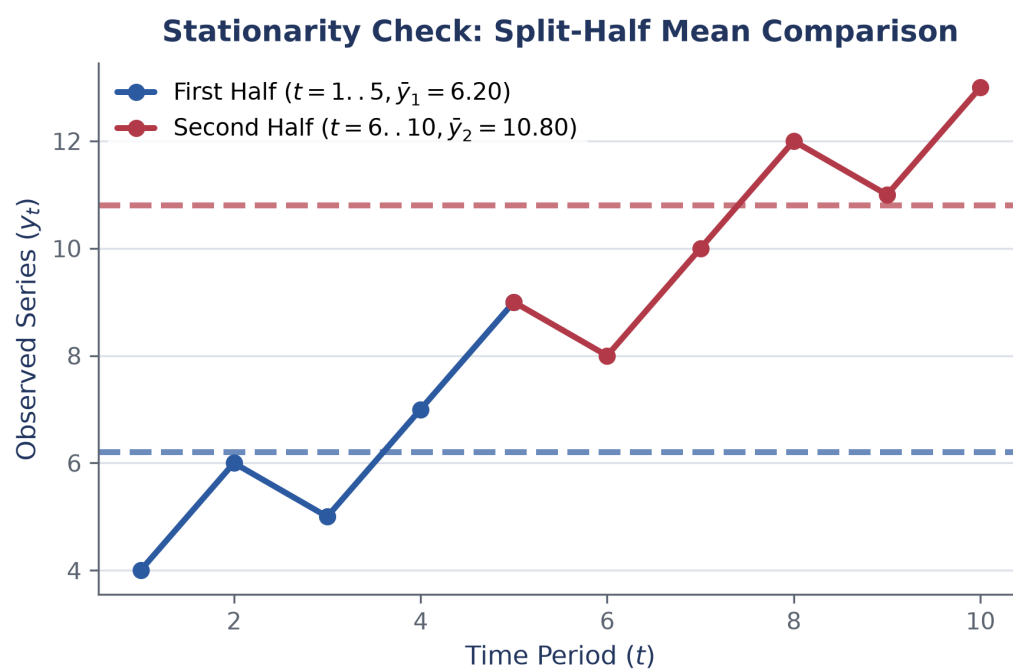


Figure 3: Split-half test comparing sub-sample means to diagnose non-stationarity.

Fully Worked Example

Given time series data ($n = 10$): $y_t = [4, 6, 5, 7, 9, 8, 10, 12, 11, 13]$.

Step 1: Split data into two equal halves: First half ($t = 1..5$): $4, 6, 5, 7, 9 \implies \bar{y}_1 = \frac{31}{5} = 6.20$. Second half ($t = 6..10$): $8, 10, 12, 11, 13 \implies \bar{y}_2 = \frac{54}{5} = 10.80$.

Step 2: Compare means: Means differ significantly ($10.80 - 6.20 = 4.60$), indicating non-stationarity due to upward trend.

Step 3: Apply first differencing $w_t = y_t - y_{t-1}$: $w_t = [6 - 4, 5 - 6, 7 - 5, 9 - 7, 8 - 9, 10 - 8, 12 - 10, 11 - 12, 13 - 11] = [2, -1, 2, 2, -1, 2, 2, -1, 2]$. New differenced mean $\bar{w} \approx 1.0$, which remains stable over time. Differencing once ($d = 1$) restores stationarity!

Watch Out! Pitfalls

Fitting standard linear regression directly on non-stationary time series leads to spurious regression with inflated R^2 values.

Key Takeaways

- Multiple regression isolates simultaneous impacts of several predictor variables.
- Time series data decomposes into Trend, Seasonal, Cyclical, and Irregular components.
- Non-stationary time series have changing means or variances over time and require differencing.